

Exercise #1

The PR robot in Fig. 1 moves on a *horizontal* plane. Its inertia matrix has the form

$$M(q) = \begin{pmatrix} A & B \cos q_2 \\ B \cos q_2 & C \end{pmatrix} > 0. \quad (1)$$

Using only the symbolic coefficients A , B and C in (1), provide the expression of the regressor matrix $Y(q, \dot{q}, \ddot{q}_r, \ddot{q}_r)$ and of the complete *adaptive* control law that guarantees global asymptotic tracking of a smooth joint trajectory $q_d(t)$, without a priori information on the values of the dynamic coefficients. Neglect all dissipative effects.

I need to compute $c(q, \dot{q})$:

$$C_1 = \frac{1}{2} \left(\frac{\partial M_1}{\partial q} + \left(\frac{\partial M_1}{\partial y} \right)^T - \frac{\partial M}{\partial q_1} \right) = \frac{1}{2} = \begin{pmatrix} 0 & 0 \\ 0 & -B s_2 \end{pmatrix} \Rightarrow c_1 = -B \dot{q}_2^2 s_2$$

$$C_2 = \frac{1}{2} \left(\frac{\partial M_2}{\partial q} + \left(\frac{\partial M_2}{\partial y} \right)^T - \frac{\partial M}{\partial q_2} \right) = 0 \Rightarrow c_2 = 0 \Rightarrow c(q, \dot{q}) = \begin{pmatrix} -B \dot{q}_2^2 s_2 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & -B s_2 \dot{q}_2 \\ 0 & 0 \end{pmatrix} \dot{q}$$

So the model is

$$\begin{cases} A \ddot{q}_1 + B C_2 \ddot{q}_2 - B s_2 \dot{q}_2^2 = \tau_1 \\ B C_2 \ddot{q}_1 + C \ddot{q}_2 = \tau_2 \end{cases} \Rightarrow \begin{cases} B(C_2 \ddot{q}_2 - s_2 \dot{q}_2^2) + A \ddot{q}_1 = \tau_1 \\ B(C_2 \ddot{q}_1) + C \ddot{q}_2 = \tau_2 \end{cases} \Rightarrow \underbrace{\begin{pmatrix} \ddot{q}_1 & C_2 \ddot{q}_2 - s_2 \dot{q}_2^2 & 0 \\ 0 & C_2 \ddot{q}_1 & \ddot{q}_2 \end{pmatrix}}_Y \begin{pmatrix} A \\ B \\ C \end{pmatrix} \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}$$

$$\Rightarrow Y(q, \dot{q}, \ddot{q}_r, \ddot{q}_r) = \begin{pmatrix} \ddot{q}_{1r} & \cos(q_2) \ddot{q}_{2r} - \sin(q_2) \dot{q}_2 \dot{q}_{2r} & 0 \\ 0 & \cos(q_2) \ddot{q}_{1r} & \ddot{q}_{2r} \end{pmatrix}$$

We set $\dot{q}_r = \dot{q}_d - \Lambda(q_d - q)$ with $\Lambda = K_D^{-1} K_P$, $K_D, K_P \succeq 0$ and we choose

the torque command:

$$\tau = \begin{pmatrix} \ddot{q}_{1r} & \cos(q_2) \ddot{q}_{2r} - \sin(q_2) \dot{q}_2 \dot{q}_{2r} & 0 \\ 0 & \cos(q_2) \ddot{q}_{1r} & \ddot{q}_{2r} \end{pmatrix} \begin{pmatrix} \hat{A} \\ \hat{B} \\ \hat{C} \end{pmatrix} + K_P e + K_D \dot{e} \quad \text{and}$$

$$\begin{pmatrix} \hat{A} \\ \hat{B} \\ \hat{C} \end{pmatrix} = \Gamma Y^T \sigma = \begin{pmatrix} \Gamma_1 & 0 & 0 \\ 0 & \Gamma_2 & 0 \\ 0 & 0 & \Gamma_3 \end{pmatrix} \begin{pmatrix} \ddot{q}_{1r} & 0 \\ C_2 \ddot{q}_{2r} - s_2 \dot{q}_2 \dot{q}_{2r} & C_2 \ddot{q}_{1r} \\ 0 & \ddot{q}_{2r} \end{pmatrix} (\dot{q}_r - \dot{q})$$

$\Gamma > 0$